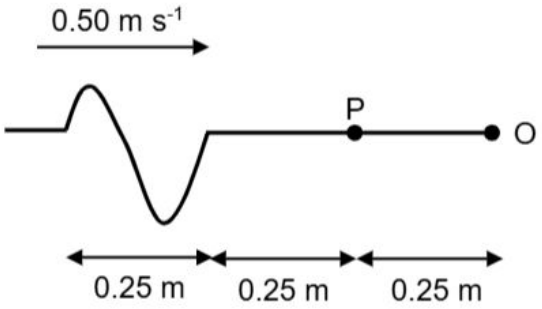
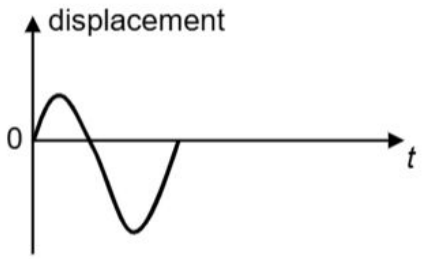
14	A wave pulse of length 0.25 m is shown at time $t = 0$. It moves along a string, fixed at point O, at 0.50 m s^{-1}. P is a point along the string 0.25 m from point O and 0.25 m from the pulse at $t = 0$.

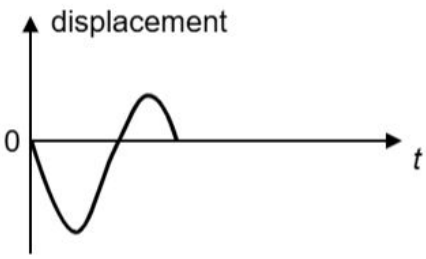


Which graph best represents the displacement of point P with time for the period t = 1.5 s to t = 2.0 s?

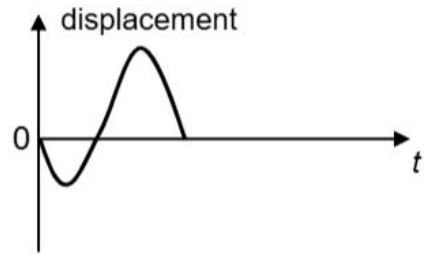
A



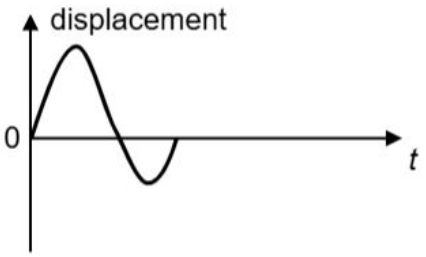
B



C



D

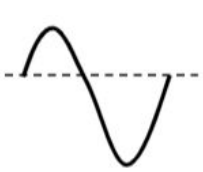


L2 **Answer: D**

Time taken for pulse to travel 0.25 m = 0.25 / 0.50 = 0.5 s

At 1.0 s, the head of the pulse just reaches O and starts to reflect.
Since O is a fixed boundary, the wave pulse undergoes 180° phase change upon reflection where its waveform is inverted as shown:

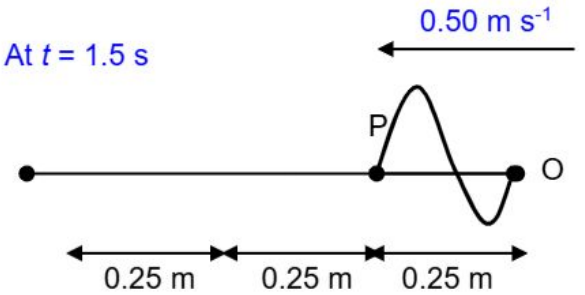
original



After 180° phase change



At 1.5 s, distance moved = 1.5 x 0.50 = 0.75 m
The head of the pulse would have moved a distance 0.75 m and reached P *after* reflection.
Between 1.5 s to 2.0 s, the pulse moves completely through P towards the left. Hence P will be deflected upwards by 'a lot' first, then downwards by 'a bit'.


0.50 m s-1
At t = 1.5 s
P
O
0.25 m
0.25 m
0.25 m